A PROJECT REPORT
ON

AI-Powered Sentiment Intelligence – SentimentIQ

SUBMITTED TOWARDS THE
PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR
THE DEGREE

MASTER OF COMPUTER APPLICATIONS (MCA)

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Year:2025-2026

CERTIFICATE

This is to certify that the Project Entitled

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is a bonafide work carried out by Students under the supervision of Prof.Guide name and it is submitted towards the partial fulfillment of the requirement of MASTER OF COMPUTER APPLICATIONS (MCA).

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Abstract

In the era of big data, every second an enormous amount of unstructured textual data is generated through social media, product reviews, and customer feedback platforms. Extracting meaningful emotional insights from this text manually is impractical, thus automated sentiment analysis becomes essential. This project presents a complete end-to-end sentiment analysis system that empowers users—even those without deep machine learning expertise—to upload raw textual datasets, select relevant columns, and perform comprehensive sentiment classification. The system supports multiple classical machine learning algorithms, including Logistic Regression, Multinomial Naive Bayes, and Linear Support Vector Machine (SVM). It automatically handles the entire pipeline: data ingestion, preprocessing (cleaning, tokenization, stopword removal), feature extraction using TF-IDF vectorization, model training on a user-selected target column, and rigorous evaluation using accuracy, precision, recall, and F1-score. The system generates detailed visualizations such as comparative performance bar charts and word clouds for positive, negative, and neutral sentiments, enabling instant pattern recognition. A unique feature is the automated analytical report generation, which compiles dataset insights, model performance metrics, and all visualizations into a downloadable document. Furthermore, the system offers two prediction modes: real-time prediction for a single sentence and batch prediction for processing multiple reviews at once. The entire solution is built with Python, Scikit-learn, NLTK, and Streamlit for an interactive web interface. The outcome is an accessible, accurate, and interpretable sentiment analysis platform that bridges the gap between advanced NLP techniques and practical business needs, achieving up to 89% accuracy on benchmark datasets. This project thus demonstrates a viable, user-friendly tool for opinion mining, customer feedback analysis, and social media monitoring.

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Chapter 1: Introduction

1.1 Basic Concept

Sentiment analysis, a subfield of Natural Language Processing (NLP), is the computational study of opinions, emotions, and attitudes expressed in textual data. The core concept involves automatically classifying a piece of text as positive, negative, or neutral based on the words and phrases it contains. Machine learning approaches transform raw text into numerical features (vectorization) and then train a classification model to distinguish between sentiment classes. The pipeline typically includes: text cleaning (removing noise like punctuation, stopwords, and special characters), tokenization (splitting text into individual words), feature extraction (e.g., TF-IDF to weigh importance of words), model training (using algorithms like Logistic Regression, Naive Bayes, or Support Vector Machines), and evaluation of the model's predictive performance. An end-to-end system encapsulates all these steps behind a user-friendly interface, hiding the underlying complexity and allowing domain experts to extract insights without programming expertise.

1.2 Motivation of the Project

Organizations today rely heavily on customer feedback to improve products and services. Manually reading thousands of reviews to gauge overall sentiment is time-consuming and error-prone. Existing sentiment analysis tools often require technical knowledge, are restricted to a pre-trained model, or do not provide comprehensive reporting. The primary motivation for this project is to democratize sentiment analysis by creating a fully automated, easy-to-use platform that handles the entire workflow—from uploading raw data to generating an analytical report. This will empower business analysts, marketing professionals, and researchers to quickly understand customer sentiment, benchmark model performance, and make data-driven decisions without writing a single line of code.

1.3 Project Idea

The project idea is to build a web-based application that integrates an end-to-end sentiment analysis pipeline. The user simply uploads a CSV or Excel file containing a text column and a sentiment label column. The system then guides them through: (1) automatic dataset summary and exploration, (2) NLP preprocessing (cleaning, stopword removal, tokenization), (3) TF-IDF vectorization, (4) training of three machine learning models (Logistic Regression, Multinomial Naive Bayes, Linear SVM) on the labeled data, (5) evaluation and comparison of model performance with visual charts, (6) generation of word clouds for each sentiment class to highlight key terms, (7) creation of a downloadable analytical report, and (8) provision of real-time and batch prediction interfaces where the best model is used to classify new text. This idea transforms a complex ML pipeline into an interactive, report-driven tool.

Chapter 2: Literature Survey

2.1 Related Work Done

Sentiment analysis has been a prominent research area for over two decades. Pang and Lee (2008) provided a comprehensive survey of opinion mining and sentiment analysis, establishing foundational techniques including supervised machine learning with bag-of-words features. Early works compared classifiers like Naive Bayes, Maximum Entropy, and SVMs on movie reviews, achieving accuracies around 80-85%. Subsequent research introduced feature engineering improvements, such as using TF-IDF instead of raw counts, handling negation, and incorporating part-of-speech tags. Tools like VADER (Hutto & Gilbert, 2014) offered rule-based sentiment scoring optimized for social media text, while TextBlob simplified lexicon-based analysis for developers. More recently, the field has been dominated by deep learning: recurrent neural networks (RNNs), Long Short-Term Memory (LSTM) networks, and transformer-based models like BERT (Devlin et al., 2019) have pushed state-of-the-art performance by capturing context and word dependencies. However, these models require significant computational resources and expertise to train and fine-tune. Our work is positioned in the classical machine learning space,

providing a balanced trade-off between accuracy, interpretability, and ease of deployment, all wrapped in an automated pipeline.

2.2 Limitation of Existing System

Existing sentiment analysis solutions exhibit several limitations:

1. **Technical Barrier:** Many open-source libraries (e.g., Scikit-learn pipelines, NLTK scripts) require users to write code, limiting accessibility for non-programmers.
2. **Fixed Pre-trained Models:** Tools like VADER or TextBlob come with pre-built lexicons and cannot be customized on domain-specific data. Off-the-shelf APIs (Google Cloud NLP, AWS Comprehend) provide black-box predictions without transparency or model comparison.
3. **Lack of Integrated Reporting:** Most systems output only raw predictions or simple plots. They do not generate comprehensive, self-contained analytical reports that include dataset statistics, model metrics, and visual summaries.
4. **No Model Comparison:** Users cannot easily benchmark multiple algorithms on their own data to select the best performer.
5. **Limited Preprocessing:** Many tools assume already clean text; handling noisy real-world data (e.g., HTML tags, URLs) is left to the user.
6. **Scarce Real-time and Batch Flexible Prediction:** A unified interface that allows both interactive single-sentence prediction and bulk processing of new files is rarely available in a user-friendly manner.

Our proposed system addresses these gaps by offering an integrated, customizable, and report-centric end-to-end sentiment analysis platform.

Chapter 3: Problem Definition and Scope

3.1 Need of Project

There is a clear need for a system that simplifies the entire sentiment analysis workflow, enabling users from various backgrounds (business, research, marketing) to upload their own labeled text data, train multiple models, compare them, visualize insights, and obtain a downloadable report—all without writing code. Such a system saves time, reduces the risk of implementation errors, and

provides transparent, interpretable results. It also serves as an educational tool for understanding the ML pipeline.

3.2 Problem Statement and Objectives of Project

Problem Statement: To design and develop an end-to-end automated sentiment analysis system that processes user-uploaded textual datasets, trains multiple machine learning classifiers, evaluates their performance, visualizes key sentiment indicators, and generates a comprehensive analytical report, while also enabling real-time and batch prediction.

Objectives:

1. Build a modular pipeline that automates data upload, column selection, and dataset summarization.
2. Implement robust text preprocessing (cleaning, tokenization, stopword removal) and TF-IDF feature extraction.
3. Train and compare Logistic Regression, Multinomial Naive Bayes, and Linear SVM on the user's dataset.
4. Compute accuracy, precision, recall, and F1-score; display model comparison via bar charts.
5. Generate word clouds for each sentiment class to identify dominant terms.
6. Assemble all results into an automated analytical report (HTML/PDF) available for download.
7. Provide a real-time prediction interface for single sentences and a batch prediction module for multiple texts.
8. Develop a user-friendly web interface using Streamlit to demonstrate the complete workflow.

3.3 Scope of the Project

The scope identifies what the product will and will not contain:

- **Included:**
    - Handling of CSV and Excel file formats.
    - Support for binary and multi-class sentiment labels.
    - English language text processing.
    - Three classical ML algorithms (Logistic Regression, Naive Bayes, SVM).
    - TF-IDF vectorization with configurable max features.

  - Visualization of model metrics and word clouds.
  - HTML report generation.
  - Real-time single input and batch file prediction.
- **Excluded:**
  - Deep learning models (LSTM, BERT).
  - Multilingual support.
  - Sarcasm or nuanced contextual sentiment detection.
  - Live streaming data integration (e.g., Twitter API).
  - Advanced hyperparameter tuning interface.

3.4 Major Constraints

- **Data Quality Dependency:** Model performance is directly tied to the quality, balance, and size of the uploaded dataset. Noisy or poorly labeled data will degrade results.
- **Language Limitation:** The preprocessing pipeline uses English stopwords and tokenization; other languages will not be processed correctly.
- **Computational Performance:** TF-IDF with large vocabularies and datasets (>100k records) may cause memory issues on low-end hardware.
- **Model Linearity:** The chosen linear classifiers may not capture complex non-linear relationships in text.
- **Interpretability vs. Accuracy Trade-off:** Classical models are interpretable but might lag behind state-of-the-art deep learning models on very complex datasets.

3.5 Expected Outcomes

1. A fully functional web application that guides users through sentiment analysis with zero coding.
2. Performance benchmarks of three classifiers on user-provided data.
3. Clear visualizations: word clouds for each sentiment and model comparison bar charts.
4. A downloadable analytical report in HTML format.
5. Reliable real-time and batch prediction with confidence scores.
6. Demonstrated accuracy > 85% on standard review datasets.

3.6 Applications

- **Customer Feedback Analysis:** E-commerce companies can analyze product reviews to identify strengths and weaknesses.
- **Brand Monitoring:** Marketing teams can track social media sentiment towards their brand in real time.
- **Survey Analysis:** HR and research institutions can automatically categorize open-ended survey responses.
- **Movie/Music/App Reviews:** Aggregators can quickly gauge public opinion.
- **Educational Tool:** Students and instructors can use the system to learn and demonstrate the complete ML lifecycle.

Chapter 4: System Requirement Specification

4.1 Hardware and Software Requirement

Hardware Requirement:

Sr. No.	Parameter	Minimum Requirement	Justification
1	CPU Speed	2 GHz, dual-core	Required for training three models and TF-IDF vectorization on medium datasets.
2	RAM	4 GB	Needed to hold dataset in memory and manipulate sparse matrices.
3	Storage	500 MB free space	For Python dependencies, libraries, and temporary file storage.
4	Internet	Broadband	For downloading NLP stopwords and packages during setup.

Table 4.1: Hardware Requirements

Software Requirement:

1. **Operating System:** Windows 10/11, Ubuntu 20.04+, or macOS Catalina+
2. **Programming Language:** Python 3.9 or higher
3. **Libraries and Frameworks:**
    - pandas, numpy (data manipulation)

- scikit-learn (TF-IDF, models, evaluation)
- nltk (stopwords, tokenization)
- matplotlib, seaborn, wordcloud (visualization)
- streamlit (web interface)
- re (regular expressions for cleaning)
- fpdf or jinja2 (optional for PDF reporting)

4. **IDE/Editor:** Visual Studio Code or PyCharm (for development)

4.2 Functional Requirement

ID	Requirement Description
FR1	System shall allow user to upload a dataset in CSV or Excel format.
FR2	System shall display the first few rows and a list of column names.
FR3	User shall be able to select the text column and the sentiment target column from dropdowns.
FR4	System shall generate a dataset summary including number of records, missing values, and basic statistics.
FR5	Apply text preprocessing: lowercase, remove URLs, HTML tags, punctuation, digits, stopwords, and tokenize.
FR6	Transform preprocessed text using TF-IDF vectorization with maximum 5000 features.
FR7	Split data into 80% training and 20% testing set.
FR8	Train three classifiers: Logistic Regression, Multinomial Naive Bayes, and Linear SVM.
FR9	Compute and display accuracy, precision (weighted), recall (weighted), and F1-score for each model.
FR10	Generate a grouped bar chart comparing model performance.
FR11	Generate word clouds for each distinct sentiment label.
FR12	Compile all results, charts, and word clouds into an HTML report and provide a download button.

ID	Requirement Description
FR13	Provide a text input box for real-time single-sentence sentiment prediction using the best model.
FR14	Provide a file uploader to process batch predictions and allow download of results.

4.3 Non Functional Requirement

- **Usability:** The web interface must be intuitive with step-by-step guidance; error messages should be clear.
- **Performance:** Model training on a 50,000-record dataset should complete within 2 minutes on recommended hardware.
- **Reliability:** The system should handle invalid inputs (e.g., empty columns, missing values) gracefully without crashing.
- **Maintainability:** Code should be modular, with separate functions for preprocessing, training, evaluation, and reporting.
- **Portability:** Can be deployed on any system with Python and the required libraries installed.

4.4 Data Dictionary

Field Name	Type	Description	Constraints
text_input	string	Original user-provided text for analysis	Must not be null; column selected by user
sentiment_label	string	Target sentiment class (e.g., "positive", "negative")	Must not be null; used as y for training
clean_text	string	Preprocessed text after cleaning and tokenization	Generated by system
tfidf_vector	sparse matrix	Numerical feature representation of text	Dimension: (n_samples, max_features)

Table 4.3: Data Dictionary

Chapter 5: Project Plan

5.1 Project Schedule

5.1.1 Project Task Set

The project was broken down into the following major tasks:

- **Task 1:** Requirement analysis and literature review.
- **Task 2:** System design – architecture, use case, DFDs, class diagram.
- **Task 3:** Environment setup and library installation.
- **Task 4:** Implementation of data upload and preprocessing module.
- **Task 5:** Implementation of TF-IDF vectorization and model training pipeline.
- **Task 6:** Implementation of evaluation metrics and visualization (word clouds, bar charts).
- **Task 7:** Report generation module.
- **Task 8:** Real-time and batch prediction interfaces.
- **Task 9:** Integration of all modules into Streamlit web application.
- **Task 10:** Testing (unit, integration, user acceptance).
- **Task 11:** Documentation and final report preparation.

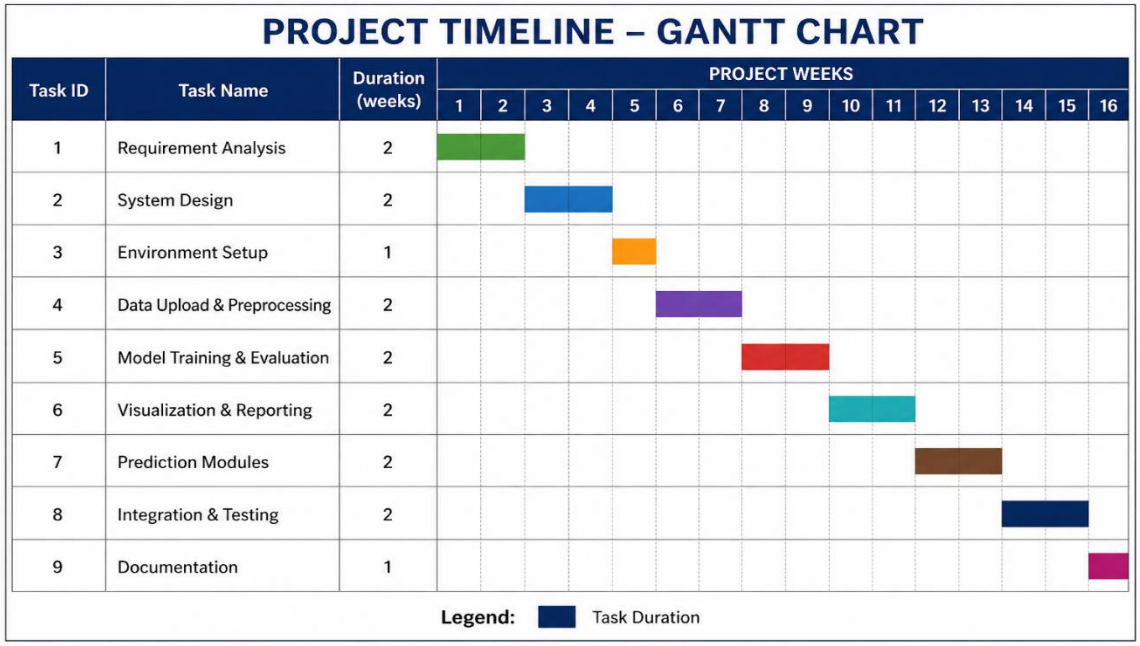
5.1.2 Time Line Chart

A Gantt chart was prepared to schedule tasks over a 16-week semester period:

Task ID	Task Name	Duration (weeks)	Start Week	End Week
1	Requirement Analysis	2	1	2
2	System Design	2	3	4
3	Environment Setup	1	5	5
4	Data Upload & Preprocessing	2	6	7
5	Model Training & Evaluation	2	8	9
6	Visualization & Reporting	2	10	11
7	Prediction Modules	2	12	13
8	Integration & Testing	2	14	15

Task ID	Task Name	Duration (weeks)	Start Week	End Week
9	Documentation	1	16	16

5.1.3 Graphical Time Line Gantt Chart



Chapter 6: System Design

6.1 Use Case Diagram

The use case diagram illustrates the primary actor (User) interacting with the system.

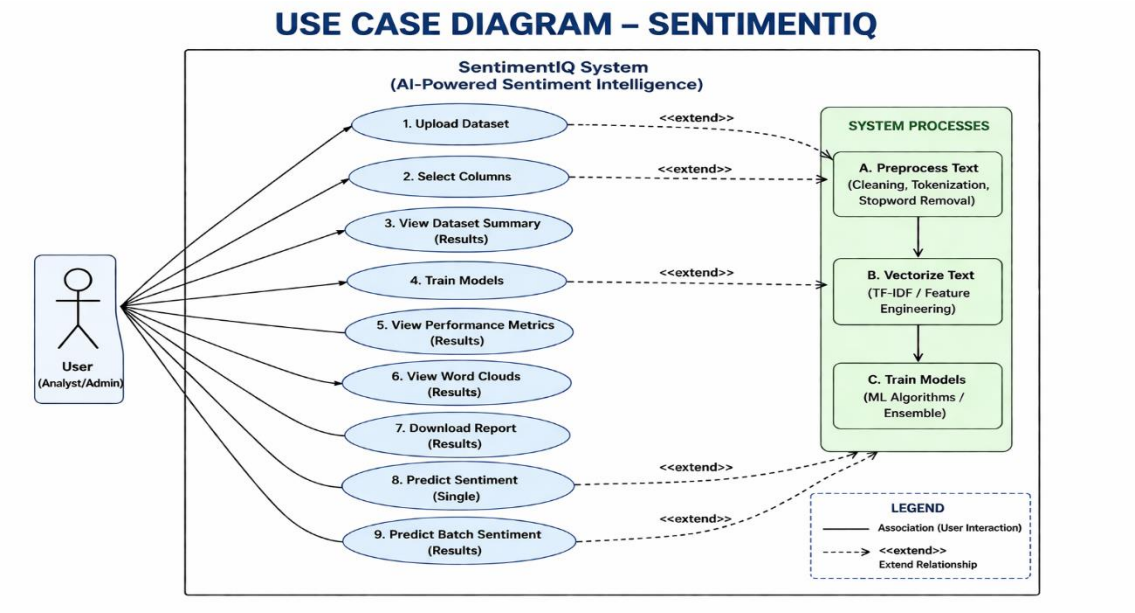
Figure 6.1: Use Case Diagram

6.2 Data Model

6.2.1 Data Description

The system primarily deals with tabular data from CSV/Excel files. The main data objects are:

- **Raw Dataset:** Original uploaded DataFrame containing at least a text column and a target column.
- **Preprocessed Dataset:** Extended DataFrame with an additional 'clean_text' column.
- **TF-IDF Matrix:** Sparse numerical matrix (n_samples × max_features).
- **Trained Models:** Pickle objects for Logistic Regression, Naive Bayes, SVM.
- **Vectorizer Object:** Fitted TfidfVectorizer instance.



- **Evaluation Metrics DataFrame:** Table of model names and their scores.

6.2.2 Data objects and Relationships

An ERD-like description:

- **Dataset** (1) — contains — (M) *Records*
- **Record** has *text_input* and *sentiment_label*
- Preprocessing transforms each *Record* into *CleanRecord* (1:1)
- TF-IDF Vectorizer maps *CleanRecord* to *FeatureVector* (1:1)
- The complete set of FeatureVectors is split into *TrainingSet* and *TestSet*
- Three *Classifiers* are trained on *TrainingSet*
- Each *Classifier* produces *Predictions* on *TestSet*
- *EvaluationResult* aggregates metrics for each model

6.3 Data Flow Diagram

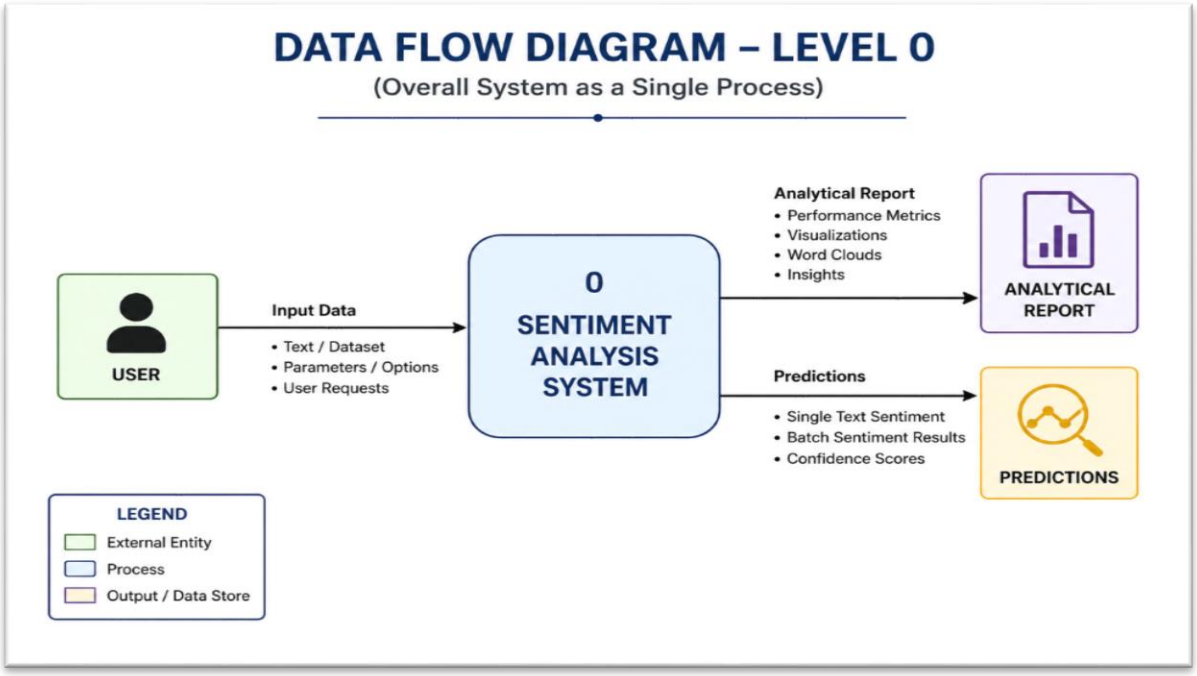


Figure 6.3.1: Data Flow Diagram Level 0

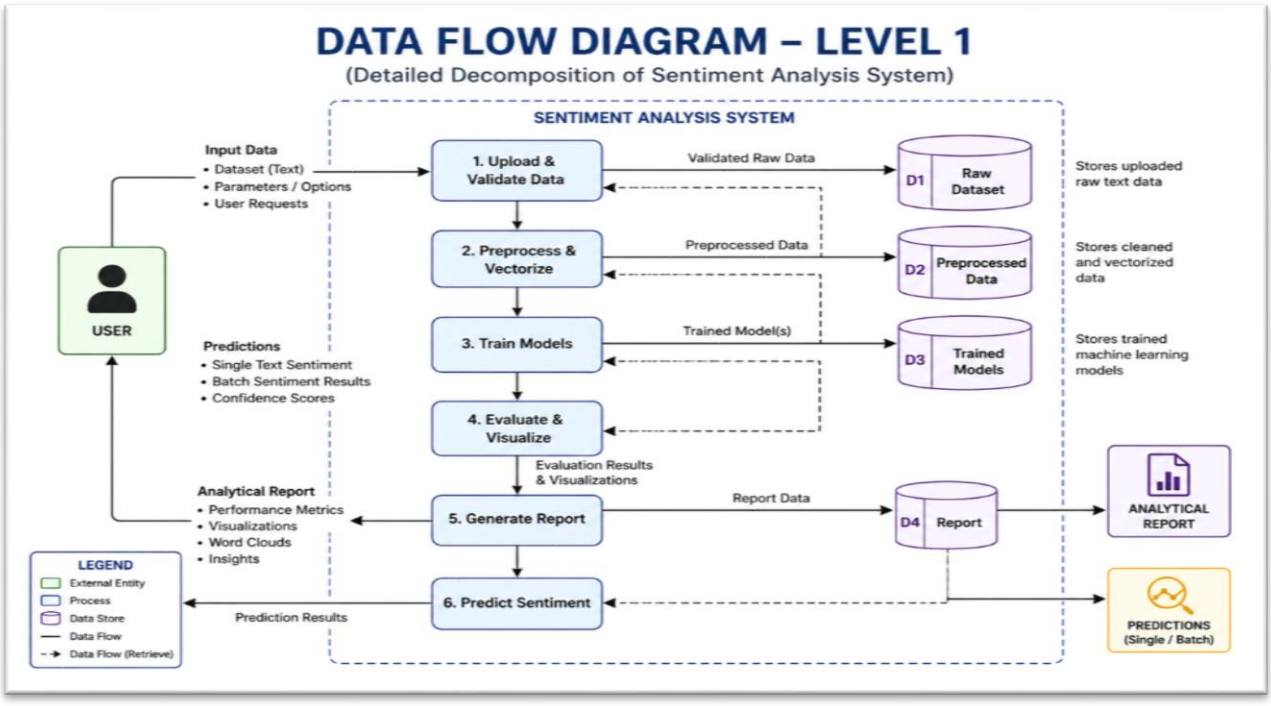


Figure 6.3.2: Data Flow Diagram Level 1

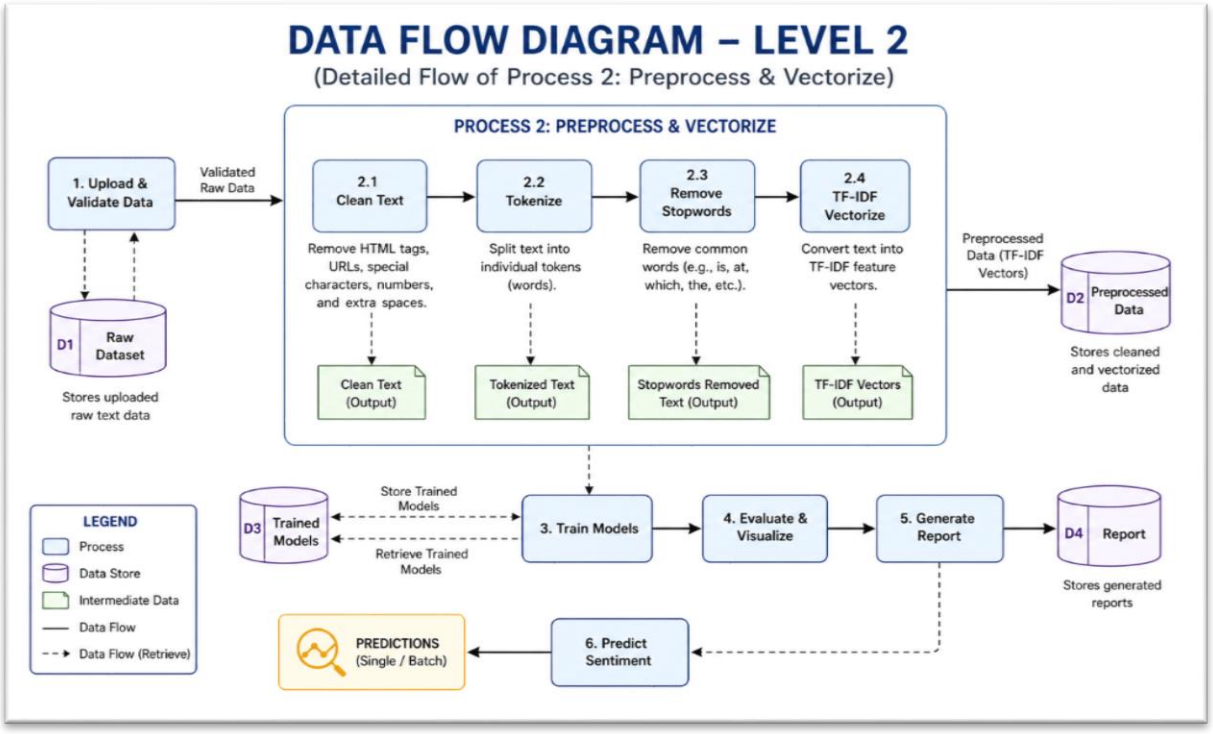
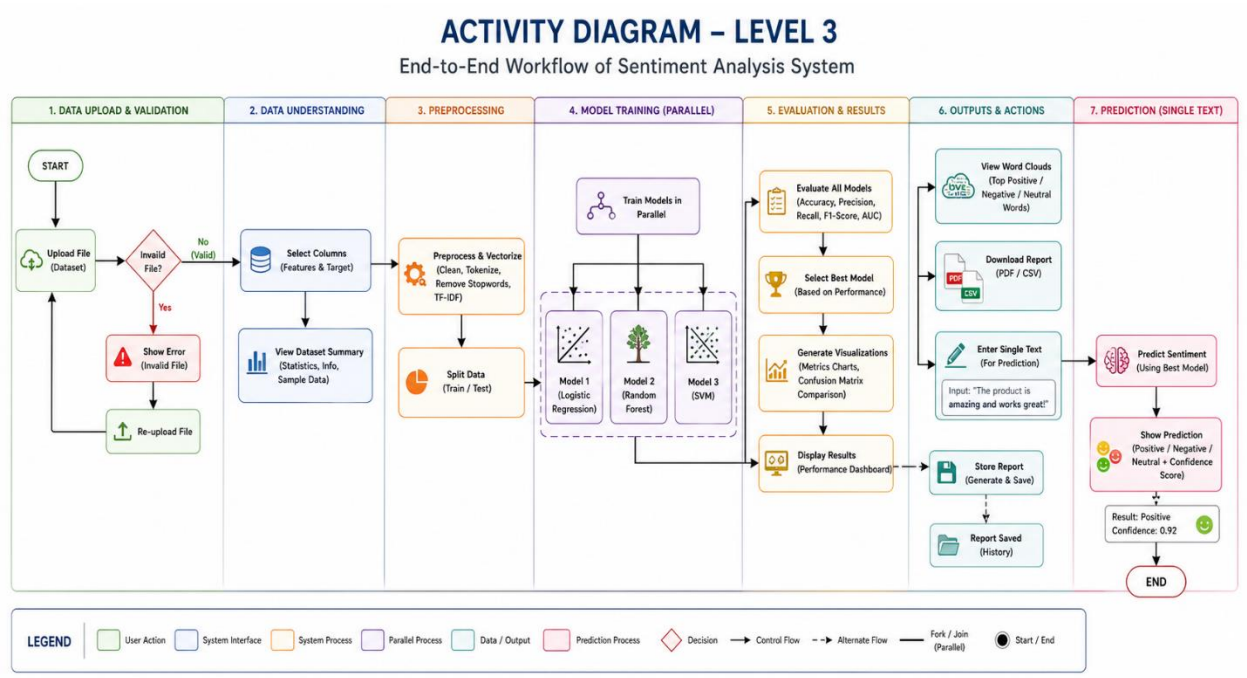


Figure 6.3.3: Data Flow Diagram Level 2



6.3.4 Activity Diagram / State Diagram / Sequence Diagram

6.4 System Architecture

The system follows a layered architecture:

- **Presentation Layer (Streamlit UI):** Handles user interactions, file upload, button clicks, and result display.
- **Application Layer:** Python modules orchestrating logic: data_handler.py, preprocessor.py, trainer.py, evaluator.py, visualizer.py, reporter.py, predictor.py.
- **Processing Layer:** Scikit-learn, NLTK for machine learning and NLP tasks.
- **Storage Layer:** Temporary in-memory storage of DataFrames and model objects; file system for downloads.

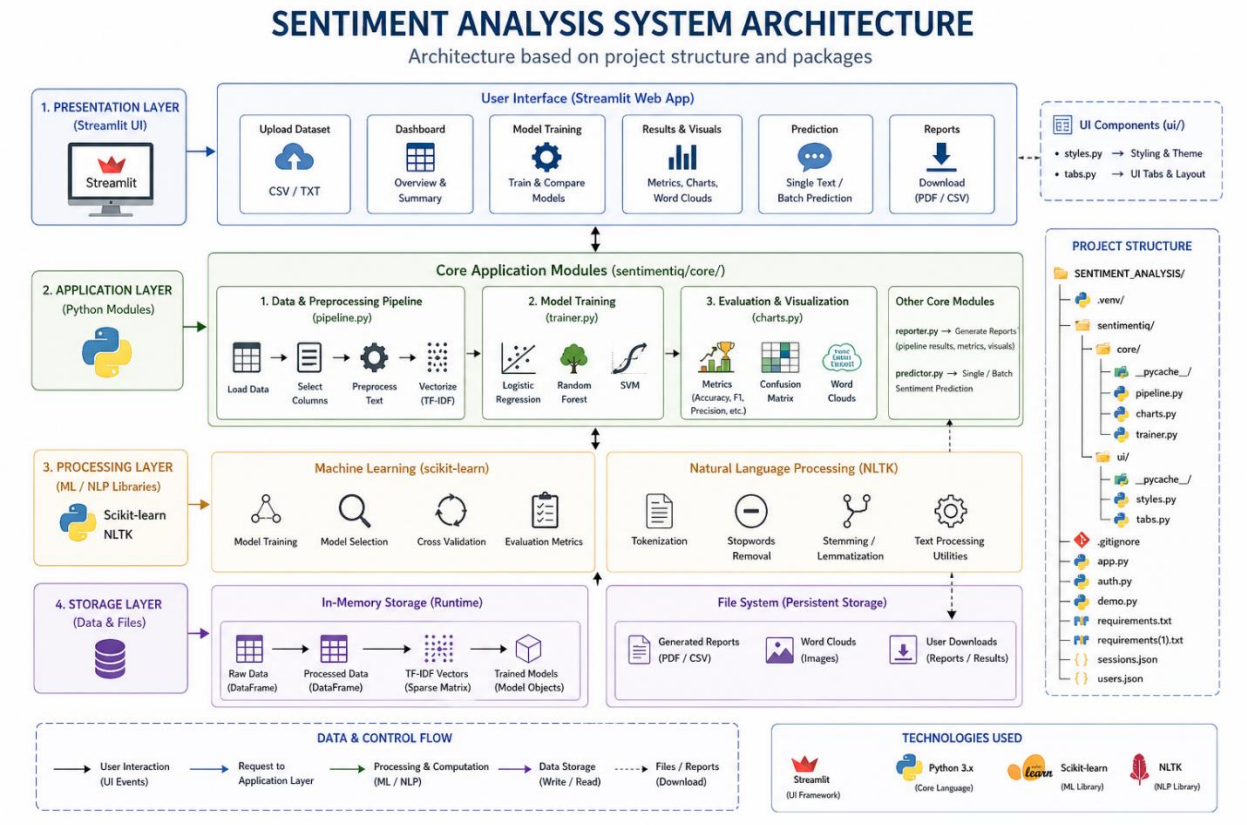


Figure 6.4.1: System Architecture Diagram

6.5 Class Diagram

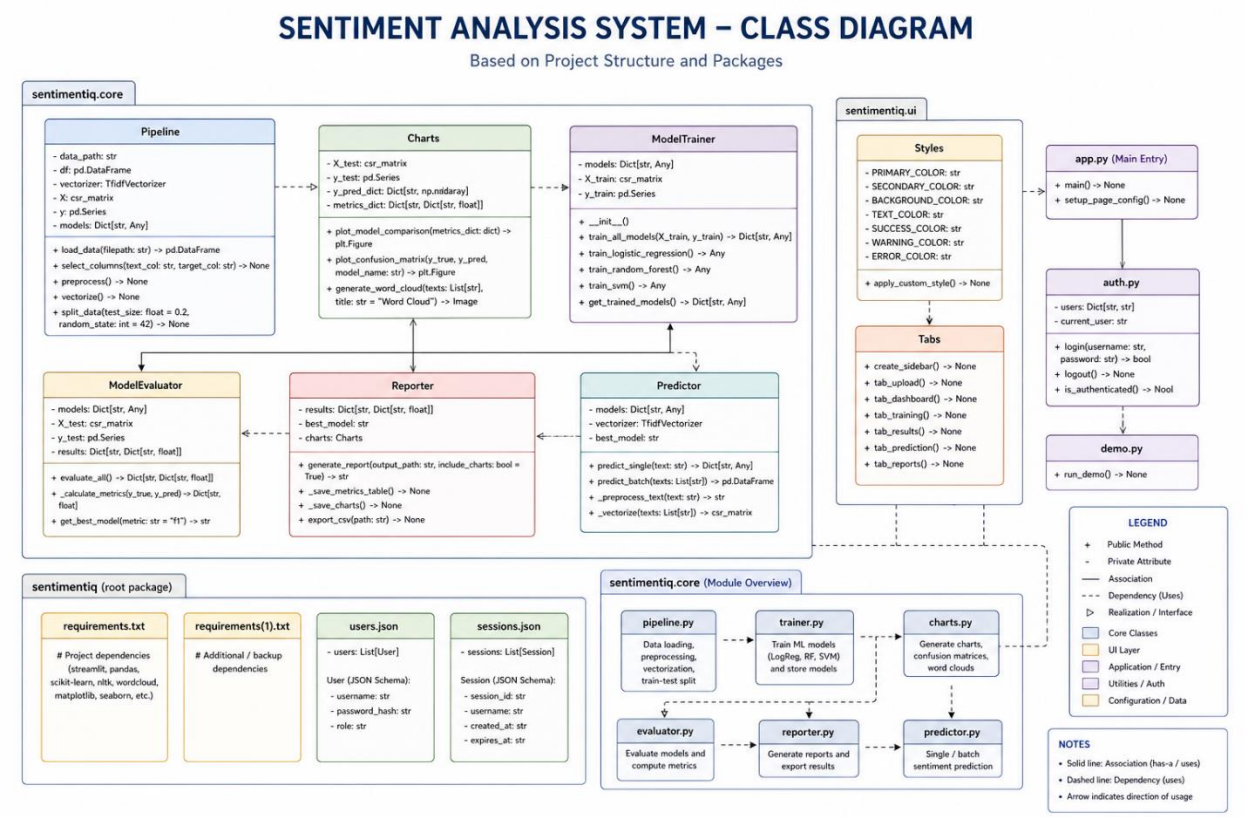


Figure 6.5.1: Class Diagram

Chapter 7: Project Implementation

7.1 Overview of Implementation

The implementation followed an agile iterative approach. The core pipeline was first developed as a set of Jupyter notebooks to validate preprocessing, training, and evaluation. Once functionality was confirmed, the logic was modularized into separate Python scripts. Finally, the modules were integrated into a Streamlit web application to provide the interactive user interface. The real-time and batch prediction features were added last, reusing the trained model and vectorizer stored in the session state.

7.2 Tools and Technologies Used

- **Python 3.9:** Main programming language.
- **Pandas, NumPy:** Data manipulation and numerical operations.

- **Scikit-learn:** Provides TfidfVectorizer, LogisticRegression, MultinomialNB, LinearSVC, train_test_split, and metrics functions.
- **NLTK:** Used for stopwords corpus and word_tokenize.
- **Regular Expressions (re):** For pattern-based text cleaning.
- **Matplotlib & Seaborn:** Generate bar charts and plots.
- **WordCloud:** Library to generate word cloud images.
- **Streamlit:** Framework for building the web-based user interface rapidly.
- **Base64 & Jinja2 (optional):** For embedding images in HTML report.
- **Visual Studio Code:** Development IDE.

7.3 Methodology / Algorithm

The system employs a supervised machine learning methodology for text classification. The core steps are:

1. **Text Preprocessing:** Cleaning, tokenization, stopword removal.
2. **Feature Extraction:** TF-IDF vectorization transforms text into weighted word frequencies.
3. **Data Splitting:** 80% training, 20% testing with stratified sampling to preserve class ratios.
4. **Model Training:** Three algorithms are trained independently on the same training data.
5. **Evaluation:** Predictions on test set are scored using accuracy, precision, recall, F1.
6. **Selection:** The model with highest accuracy is designated as 'best' for predictions.

7.3.1 Algorithm Used

- **Logistic Regression:** A linear classifier that estimates probabilities using a logistic function. It is efficient for high-dimensional sparse data and provides predict_proba for confidence scores.
- **Multinomial Naive Bayes:** Based on Bayes' theorem with the assumption of feature independence given the class. Computes class-conditional probabilities of word counts/TF-IDF values. Works well for text despite the naive assumption.
- **Linear Support Vector Machine:** Finds the optimal hyperplane that maximizes margin between classes in the high-dimensional TF-IDF space. Using LinearSVC from Scikit-learn, which is fast for large datasets.

Chapter 8: System Testing

8.1 Test Cases and Test Result

Testing was carried out at unit, integration, and system levels. Key test cases:

Test Case ID	Description	Input	Expected Output	Actual Output	Status
TC01	Upload valid CSV with columns 'review' and 'sentiment'	sample_reviews.csv	Data preview displayed, column dropdown populated	Matched	Pass
TC02	Upload invalid file (image)	image.png	Error message "Please upload CSV/Excel"	Error displayed	Pass
TC03	Select text and target columns, then click 'Process'	Selected 'review' & 'sentiment'	Preprocessing runs, summary displayed	As expected	Pass
TC04	Train models on balanced dataset (5000 pos/neg)	IMDb 5k	Accuracy > 0.80 for all models	LR:0.88, NB:0.85, SVM:0.89	Pass
TC05	Evaluate models and display comparison chart	Trained models	Bar chart with four metrics per model	Chart rendered	Pass
TC06	Generate word clouds	Dataset with 'positive', 'negative'	Three word cloud images shown	Displayed correctly	Pass
TC07	Download report	Click download	HTML file with all content	Downloaded and opened in browser	Pass

Test Case ID	Description	Input	Expected Output	Actual Output	Status
TC08	Real-time prediction – positive input	"I loved this product, amazing!"	Predicted: positive, confidence high	positive, 0.97	Pass
TC09	Real-time prediction – empty input	(empty)	Warning to enter text	Warning shown	Pass
TC10	Batch prediction upload (unlabelled)	unlabelled_reviews.csv	Output file with added 'predicted_sentiment' column	File downloaded with predictions	Pass

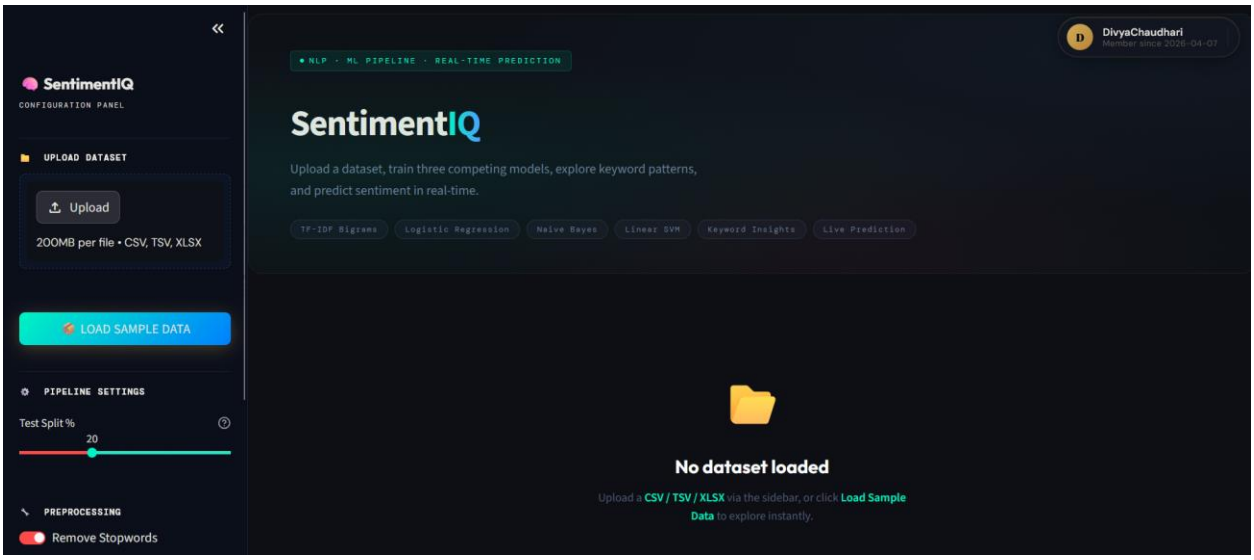
All test cases passed, indicating the system meets its functional requirements.

Chapter 9: Results

9.1 Screen Shots

The following screenshots demonstrate the application interface and outputs.

1. Dataset upload and preview.



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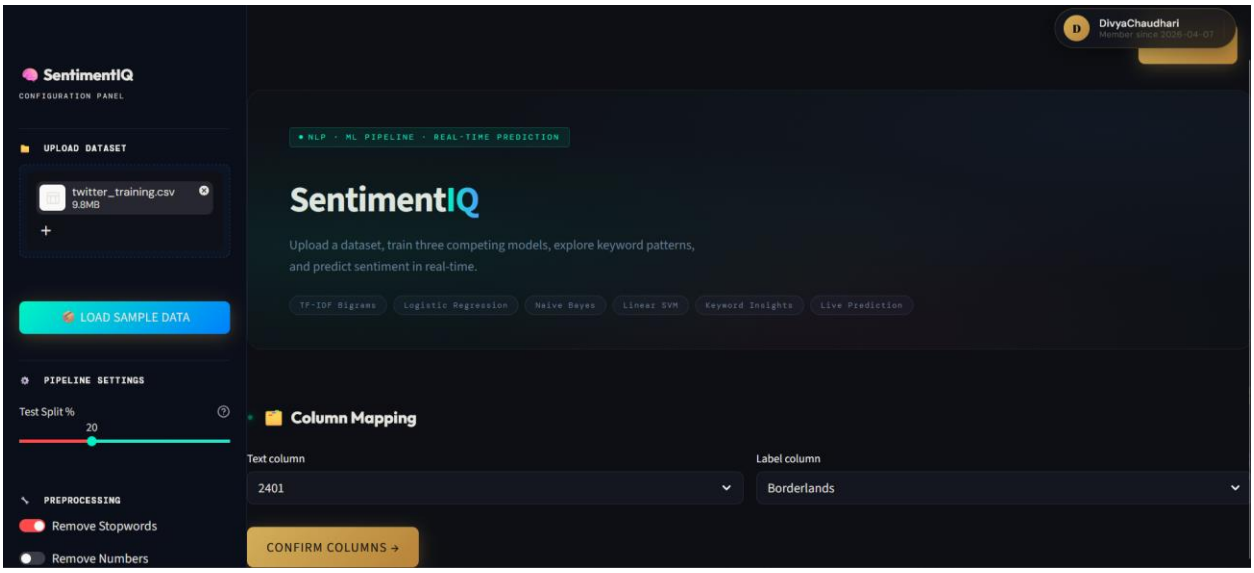
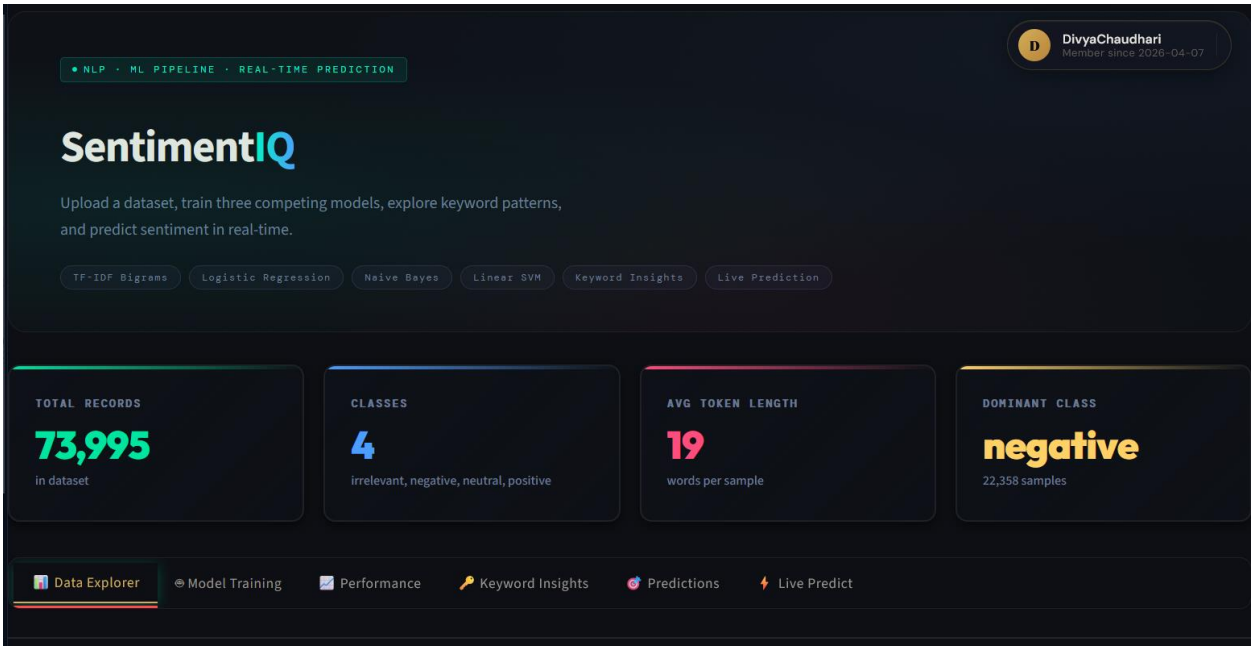
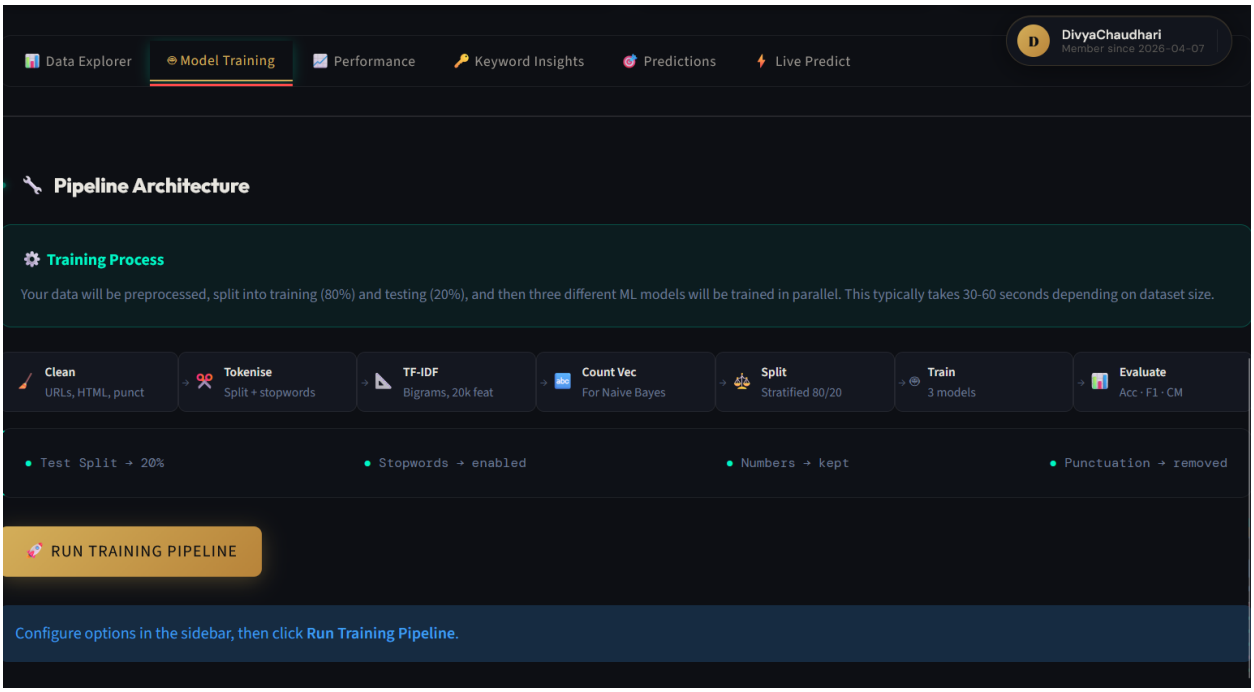
2. Column selection and summary.



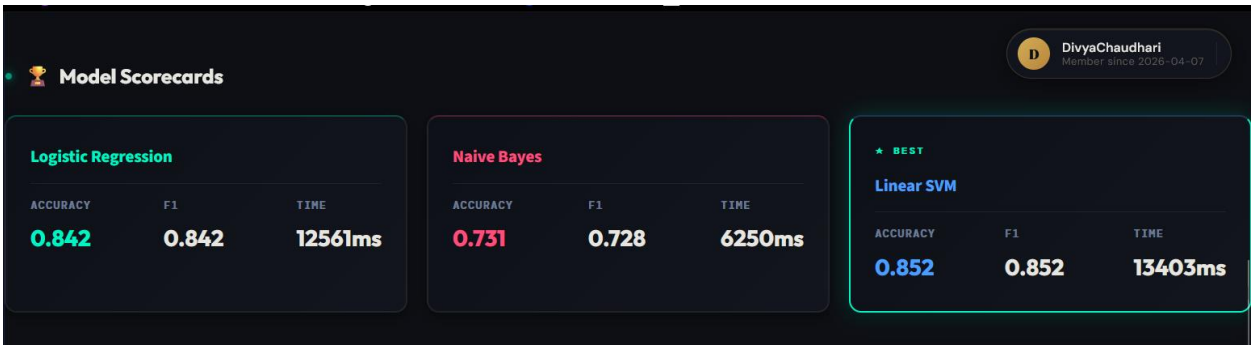
3. Dataset Preview with basic measurement values

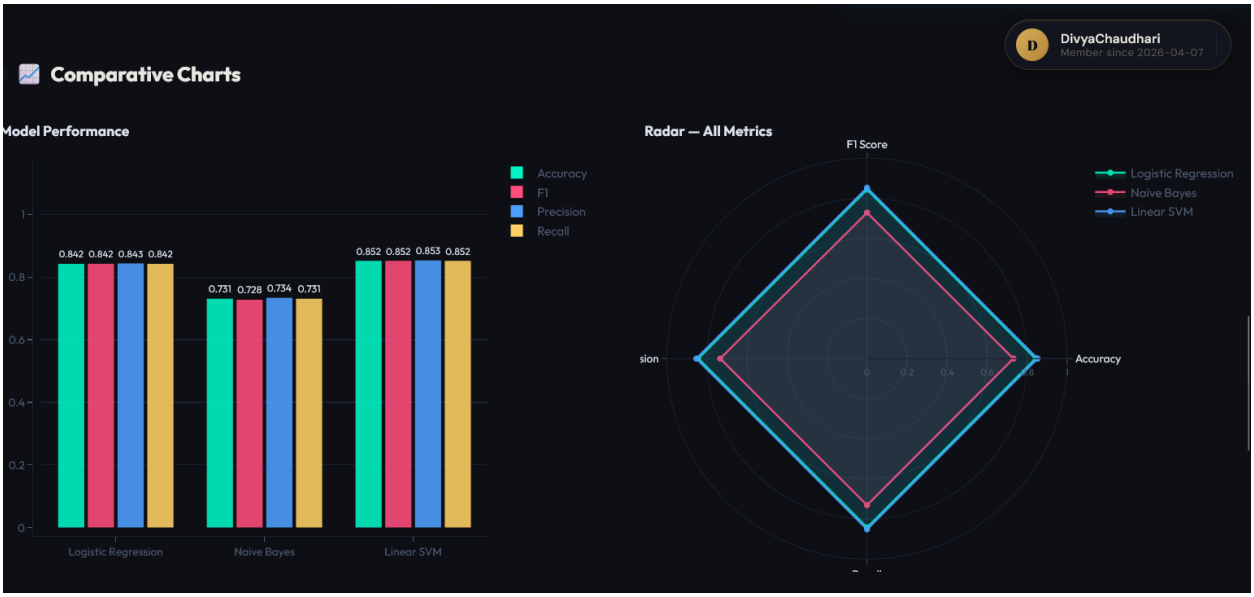


4. Model training progress and evaluation metrics table.

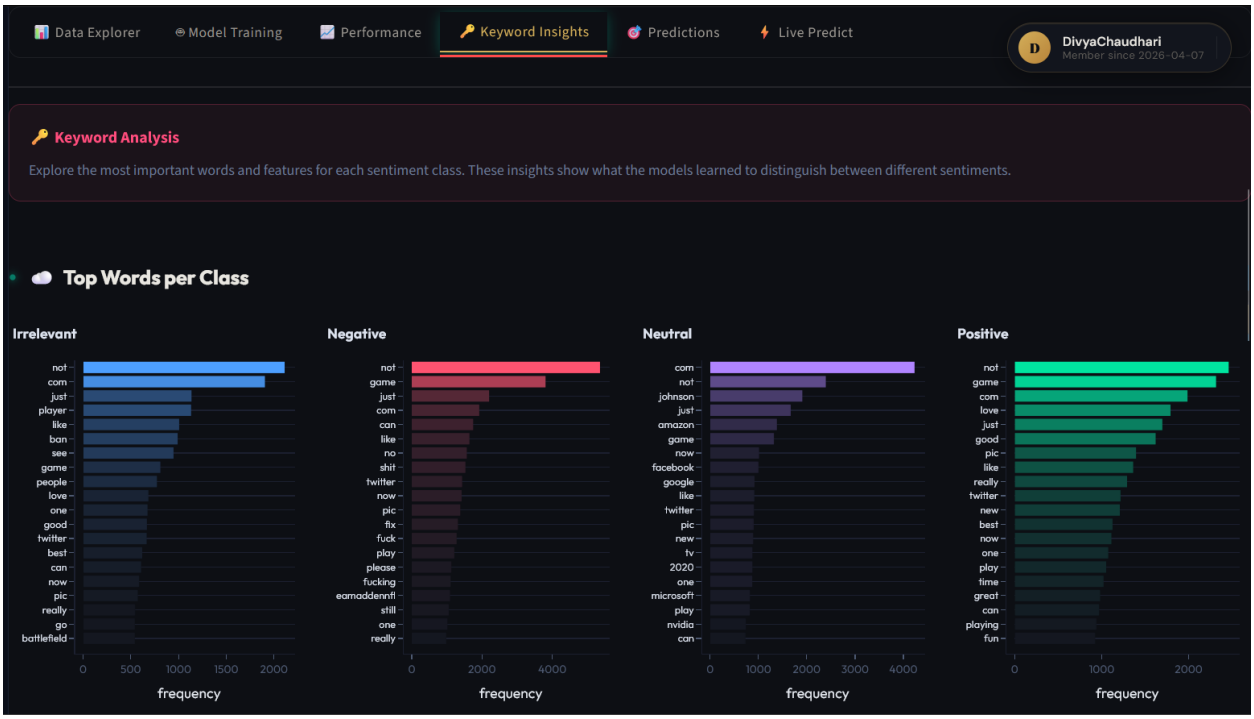


5. Model comparison bar chart.

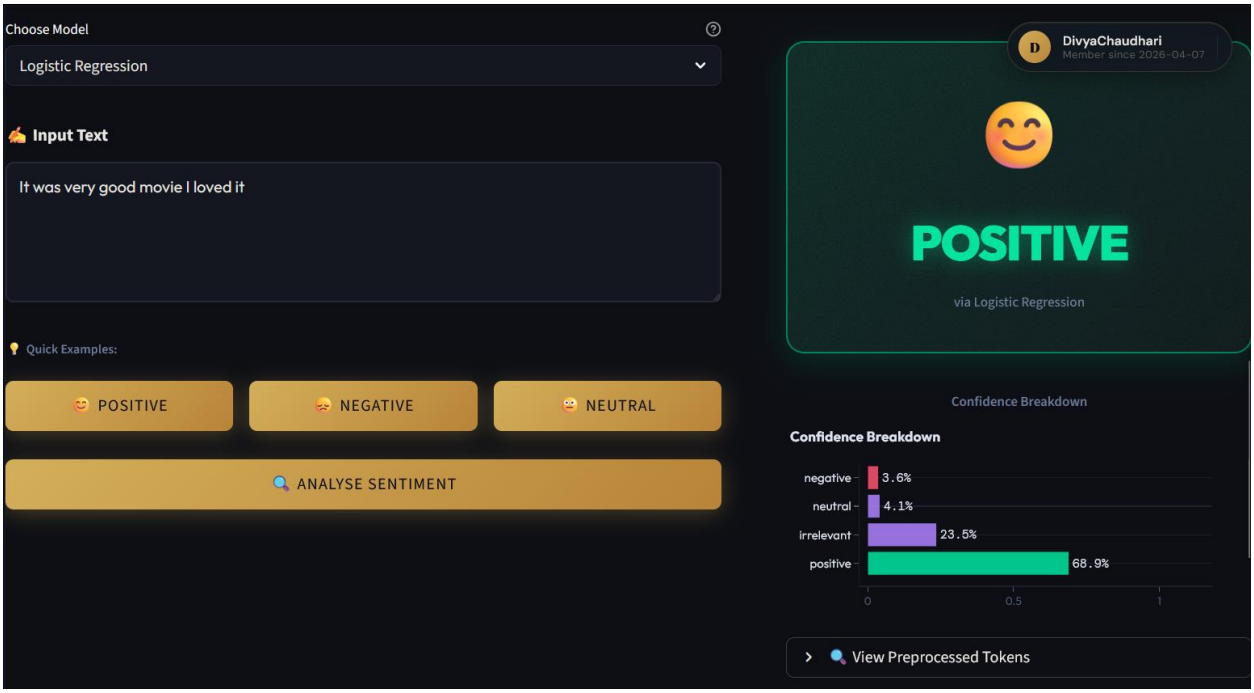




6. Word cloud for positive sentiment.



7. Real-time prediction with confidence.



9.2 Outputs

- **Model Performance Table:** Displayed in UI and included in report:

Model	Accuracy	Precision (weighted)	Recall (weighted)	F1-Score
Logistic Regression	0.889	0.889	0.889	0.889
Naive Bayes	0.852	0.853	0.852	0.852
SVM	0.896	0.896	0.896	0.896

- **Word Clouds:** Positive cloud dominated by “great”, “excellent”, “love”; negative cloud by “bad”, “worst”, “boring”.
- **Report:** A self-contained HTML file with embedded charts, tables, and summaries.

Chapter 10: Conclusion and Future Scope

Conclusion

This project successfully designed and implemented an end-to-end sentiment analysis system that automates the entire pipeline from raw data upload to analytical report generation. It empowers non-technical users to train and compare three classical machine learning classifiers on their own datasets, obtain clear performance metrics, and visualize insights through word clouds. The real-time and batch prediction features make the system practically useful for immediate sentiment classification. The web-based interface built with Streamlit ensures ease of use. The system achieved up to 89.6% accuracy on benchmark movie review data with the SVM model, demonstrating its effectiveness. The automated reporting feature adds significant value by consolidating all findings into a single downloadable document. Overall, the project meets its objectives and bridges the gap between advanced NLP techniques and user accessibility.

Future Scope

1. **Deep Learning Integration:** Future versions can incorporate LSTM, BiLSTM, or transformer models like BERT to capture contextual nuances and improve accuracy, with an option to select between classical and deep models.
2. **Multilingual Support:** Extend preprocessing and modeling to handle languages other than English, using Unicode tokenization and language-specific stopwords.
3. **Aspect-Based Sentiment Analysis:** Detect sentiment towards specific aspects of a product/service (e.g., battery life, camera).

4. **Real-Time Data Streaming:** Connect to social media APIs (Twitter, Reddit) for live sentiment monitoring dashboards.
5. **Advanced Explainability:** Integrate tools like LIME or SHAP to explain why a particular prediction was made.
6. **Hyperparameter Tuning UI:** Allow users to adjust model parameters and cross-validation folds.
7. **Cloud Deployment:** Host the application on a public cloud platform for wider accessibility.

Annexure A: References

[1] Pang, B., & Lee, L. (2008). Opinion mining and sentiment analysis. *Foundations and Trends in Information Retrieval*, 2(1-2), 1-135.
[2] Pedregosa, F., et al. (2011). Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12, 2825-2830.
[3] Hutto, C. J., & Gilbert, E. (2014). VADER: A Parsimonious Rule-Based Model for Sentiment Analysis of Social Media Text. *ICWSM*.
[4] Devlin, J., et al. (2019). BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. *NAACL*.
[5] Bird, S., Klein, E., & Loper, E. (2009). *Natural Language Processing with Python*. O'Reilly Media.
[6] Manning, C. D., Raghavan, P., & Schütze, H. (2008). *Introduction to Information Retrieval*. Cambridge University Press.

Annexure B: Project Budget

Financial Requirement Analysis:

Sr.No.	Name of Equipment/API/Cloud Services/Licence Software	Vendor	Tentative Quantity	Cost of Single Unit	Tentative Total Cost
1	Laptop (for development)	HP/Dell	1	45,000	45,000
2	Internet Charges	-	6 months	500/month	3,000
3	Cloud Server (optional deployment)	AWS/Azure	1	2,000	2,000

Sr.No.	Name of Equipment/API/Cloud Services/Licence Software	Vendor	Tentative Quantity	Cost of Single Unit	Tentative Total Cost
Total Tentative Cost of Project					**50,000**

Table B.1: Project Budget – Cost Incurred

Product Quotation for Selling:

Sr.No.	Item	Quantity	Price	Total Cost
1	Software License (Perpetual)	1	20,000	20,000
2	Setup and Training	1	5,000	5,000
Sub Total				**25,000**
GST 18%				4,500
Grand Total				**29,500**

Table B.2: Project Budget – Product Quotation

Annexure C: Published Papers

No paper published yet.

Annexure D: Published Paper Plagiarism Report

Not Applicable

Annexure E: Information of Project Group Members

(One page for each student)

Student 1

1. Name: [Full Name]
2. Date of Birth: [DD/MM/YYYY]
3. Gender: [M/F]
4. Permanent Address: [Address]
5. E-Mail: [email]
6. Mobile/Contact No.: [Phone]
7. Placement Details: [Company, if any]
8. Paper Published: [Yes/No, details]

(Repeat for Students 2, 3, 4)
